Trisomy 4+10 Status and Survival Outcomes in Pediatric B-Cell Acute Lymphoblastic Leukemia

BINF 6310 — Advanced Statistics for Genomics

Abstract

Trisomy of chromosomes 4 and 10 (trisomy 4+10) is a well-established favorable cytogenetic marker in pediatric B-cell acute lymphoblastic leukemia (B-ALL). This study applies a Welch two-sample t-test to evaluate whether trisomy 4+10 status is associated with differences in survival duration among deceased patients in the TARGET-ALL-P2 cohort ($n = 254$ deceased). Trisomy 4+10 positive patients had a mean survival of 1,091.6 days compared to 1,112.1 days in the negative group. The difference was not statistically significant ($t = -0.188$, $p = 0.851$), a result interpreted in the context of survivorship bias inherent to restricting analysis to deceased patients only.

Introduction

Pediatric B-ALL is the most common childhood malignancy, and cytogenetic risk stratification is central to treatment planning. Simultaneous trisomy of chromosomes 4 and 10 is a favorable prognostic marker associated with improved event-free and overall survival [1]. Patients carrying this marker are frequently classified as standard risk and respond well to standard chemotherapy regimens.

The TARGET-ALL Phase 2 (TARGET-ALL-P2) dataset, available through the NCI Genomic Data Commons, provides a rich clinical and molecular resource for investigating such markers in a real-world pediatric cohort. This analysis classifies patients by trisomy 4+10 status using molecular test records and applies a Welch two-sample t-test to assess whether survival duration differs between groups among deceased patients.

Methods

Data

Data were obtained from the TARGET-ALL-P2 project via the GDC Data Portal. Two files were used: `clinical.tsv` (1,587 patients) and `follow_up.tsv`, which contains molecular test records with cytogenetic chromosome annotations (`molecular_tests.chromosome`) coded as `chr4` and `chr10`.

Patient Classification

Patients were classified as trisomy 4+10 positive if their `cases.case_id` appeared in both the `chr4` and `chr10` subsets of the molecular test records. All remaining patients were classified as negative. Of 1,587 patients, 1,027 (64.7%) were trisomy 4+10 positive and 560 (35.3%) were negative.

Outcome Variable

The primary outcome was `demographic.days_to_death`, restricted to patients with vital status recorded as "Dead" ($n = 254$; 164 positive, 90 negative). This variable represents days from index date to death and serves as a proxy for survival duration.

Statistical Analysis

A Welch two-sample t-test was used to compare mean days to death between groups. Welch's correction was chosen over Student's t-test because the groups differ substantially in size (164 vs. 90), making the assumption of equal variances unreliable. Welch's method adjusts degrees of freedom based on the observed variance of each group independently. All analyses were performed in R 4.4.2 using the `dplyr` package, with verification in Python 3 using `scipy.stats`.

Results

Of the 254 deceased patients, 164 were trisomy 4+10 positive and 90 were negative. The mortality rate was nearly identical between groups: 16.0% for positive patients (164/1,027) and 16.1% for negative patients (90/560), suggesting that trisomy 4+10 status did not strongly influence whether a patient died (Figure 3).

Table 1: Summary of days to death by trisomy 4+10 status

Group	n	Mean (days)	SD
Trisomy 4+10 Positive	164	1091.6	824.8
Trisomy 4+10 Negative	90	1112.1	829.9

The Welch two-sample t-test found no statistically significant difference in days to death between groups ($t = -0.188$, $df = 175.8$, $p = 0.851$). The distributions of days to death were highly overlapping with similar variance in both groups (Figure 2), consistent with the non-significant result. The boxplot (Figure 1) further illustrates the similarity in median survival and spread between groups.

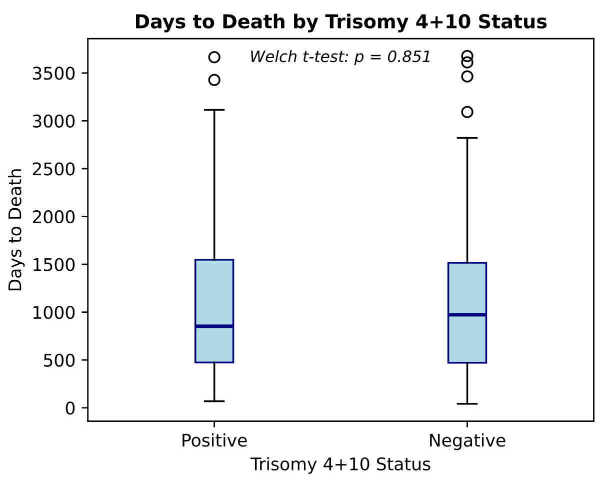

Figure 1: Boxplot of days to death by trisomy 4+10 status among deceased patients ($n = 254$). Medians and interquartile ranges are similar between groups. Welch t-test $p = 0.851$.

Discussion

The absence of a significant difference in days to death between trisomy 4+10 positive and negative patients ($p = 0.851$) is an informative finding requiring careful interpretation. This result does not contradict the established clinical literature on trisomy 4+10 as a favorable marker. Rather, it reflects a key analytical limitation: by restricting analysis to deceased patients only, we introduce survivorship bias. The benefit of trisomy 4+10 manifests primarily in *preventing* death, not in extending survival among those who do die. Patients with this marker are more likely to achieve remission and are therefore less likely to appear in the deceased subset.

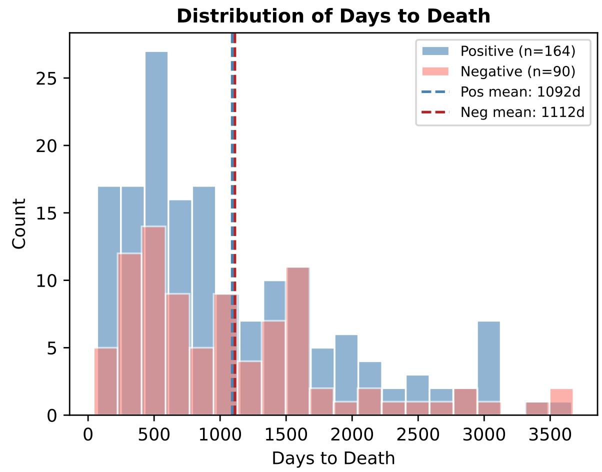

Figure 2: Overlapping histograms of days to death for trisomy 4+10 positive (blue) and negative (red) deceased patients. Dashed lines indicate group means. Distributions are broadly similar.

The nearly identical mortality rates (16.0% vs. 16.1%) further support this interpretation. Both groups lost roughly the same proportion of patients, but the absolute numbers differ substantially — 164 deaths among 1,027 positive patients vs. 90 among 560 negative patients — reflecting the larger size of the positive group rather than a differential mortality risk.

Welch's t-test was the appropriate choice given the unequal group sizes (164 vs. 90). Unlike Student's t-test, Welch's method does not assume homogeneity of variance. The standard deviations were nearly equal (824.8 vs. 829.9), confirming that either test would have yielded similar results, but Welch's remains the safer default for unbalanced designs.

Future analyses could extend this framework using Bayesian inference to incorporate prior knowledge from the literature into probability estimates of poor outcome. A beta-binomial model with an informative prior derived from published survival rates would allow uncertainty quantification beyond the binary significant/non-significant interpretation, and would be better suited to the full cohort rather than the deceased subset alone.

Conclusion

This analysis applied a Welch two-sample t-test to compare survival duration between trisomy 4+10 positive and negative deceased patients in the TARGET-ALL-P2 cohort. No statistically significant difference was found ($p = 0.851$), a result best explained by sur-

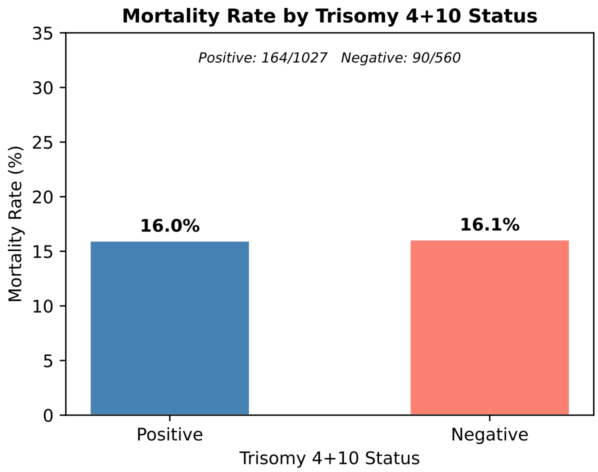

Figure 3: Mortality rate by trisomy 4+10 status across the full cohort. Rates are nearly identical (16.0% vs. 16.1%), suggesting the marker does not influence overall mortality rate in this cohort.

vivorship bias inherent to analyzing deceased patients only. The finding underscores that the clinical benefit of trisomy 4+10 lies in improving overall survival rates rather than extending survival among patients who ultimately succumb to disease.

References

[1] Borowitz, M.J., et al. (2008). Clinical significance of minimal residual disease in childhood acute lymphoblastic leukemia and its relationship to other prognostic factors. *Blood*, 111(12), 5477–5485.